CS 8674 Project Course - Proposal & Semester Plan (Syllabus)

The Role of Generative AI in Enhancing Human Communication

Design and Development of CallingIn: An AI-Assisted Intentional Communication Platform

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Table of Contents:

1. Project Overview 1
2. Research Objectives 2
3. Project Outline 3
4. System Overview "CallingIn" 4
5. Technology Stack 6
6. Semester Timeline, Syllabus 7
7. Deliverables 8
8. GitHub Repository 9
9. Final Reflection 9
10. References 9

1. Project Overview

➢ Background

Many people hesitate to begin important conversations because they are unsure how to express themselves, fear being misunderstood, or simply do not know how to start. While modern messaging platforms make communication easier, they do little to help users begin meaningful conversations.

➢ Project Description

This project explores how Generative AI can enhance human communication through the design and development of **CallingIn**, an AI-assisted communication platform that helps users initiate meaningful conversations while preserving their authentic voice.

Rather than replacing existing messaging applications, CallingIn focuses on helping users overcome the emotional barriers associated with beginning important conversations. The platform combines intentional conversation requests with AI-assisted communication coaching to guide users from the moment they decide to reach out until a meaningful conversation begins.

❖ Example Use Case.

Scenario of **Reconnecting with a Friend**

User Input: "We haven't talked in a long time and I miss our friendship, but I don't know how to reach out."

CallingIn AI Suggestion: "I've been thinking about our friendship lately and wanted to check in. I'd really appreciate the opportunity to reconnect if you're open to it."

The recipient then receives a conversation invitation and can choose:

- ❑ Accept
- ❑ Maybe Later
- ❑ Decline

Only after acceptance does a private conversation begin.

➢ Problem Statement

Current communication platforms primarily focus on delivering messages rather than helping users overcome the emotional challenges that often prevent meaningful conversations from taking place.

This project investigates how Generative AI can assist individuals in initiating thoughtful, authentic conversations while preserving their own voice and emotional intent.

2. Research Objectives

➢ Research Question

How can Generative AI assist individuals in initiating meaningful interpersonal conversations while preserving authenticity and emotional intent?

Objectives

- Study the role of Generative AI in enhancing interpersonal communication.
- Design and develop **CallingIn**, an AI-assisted communication platform.
- Evaluate how Generative AI can help users draft thoughtful and authentic messages.

- Analyze the effectiveness, benefits, and limitations of AI-assisted communication.
- Contribute practical insights into the use of Generative AI for meaningful human communication.

3. Project Outline

❑ Introduction

This project combines academic research on Generative AI with the design and implementation of a functional software application to evaluate AI-assisted communication in real-world scenarios.

- Background of Generative AI
- Communication challenges
- Importance of meaningful conversations

❑ Purpose of the Project

- Explore how AI can improve interpersonal communication.
- Design and implement CallingIn.
- Evaluate AI-assisted communication support.

❑ Scope

The project focuses on AI-assisted text communication and intentional conversation initiation.

The project does not include:

- Voice calls
- Video calls
- Group messaging
- Social media features

Literature Review

Research topics will include:

- Generative AI
- Human-computer interaction
- AI-assisted communication
- Prompt engineering
- Ethical AI
- Digital well-being

System Design

- Users
- Conversation Requests
- AI Communication Coach
- Private Conversations
- Social Battery

Implementation

- Frontend
- Backend
- Database
- Authentication
- AI Integration

Evaluation

The completed application will be evaluated based on usability, functionality, user experience, and the effectiveness of the AI Communication Coach.

Future Work

Potential future enhancements include:

- Mobile applications
- End-to-end encryption
- Additional AI communication styles
- Relationship reminders
- Personalized communication insights

4. System Overview – CallingIn

Core Features

- User Registration and Authentication
- AI Communication Coach
- Conversation Requests
- Private One-to-One Messaging
- Social Battery Status
- AI-assisted Message Rewriting
- User Profiles

The platform focuses on helping users begin meaningful conversations rather than replacing existing messaging platforms.

❖ Example User Flow

1. User signs into CallingIn.

2. User selects a person they would like to reconnect with.

3. User explains why they would like to begin the conversation.

4. The AI Communication Coach generates a thoughtful message while preserving the user's own voice.

5. The user reviews and edits the suggestion.

6. A conversation invitation is sent.

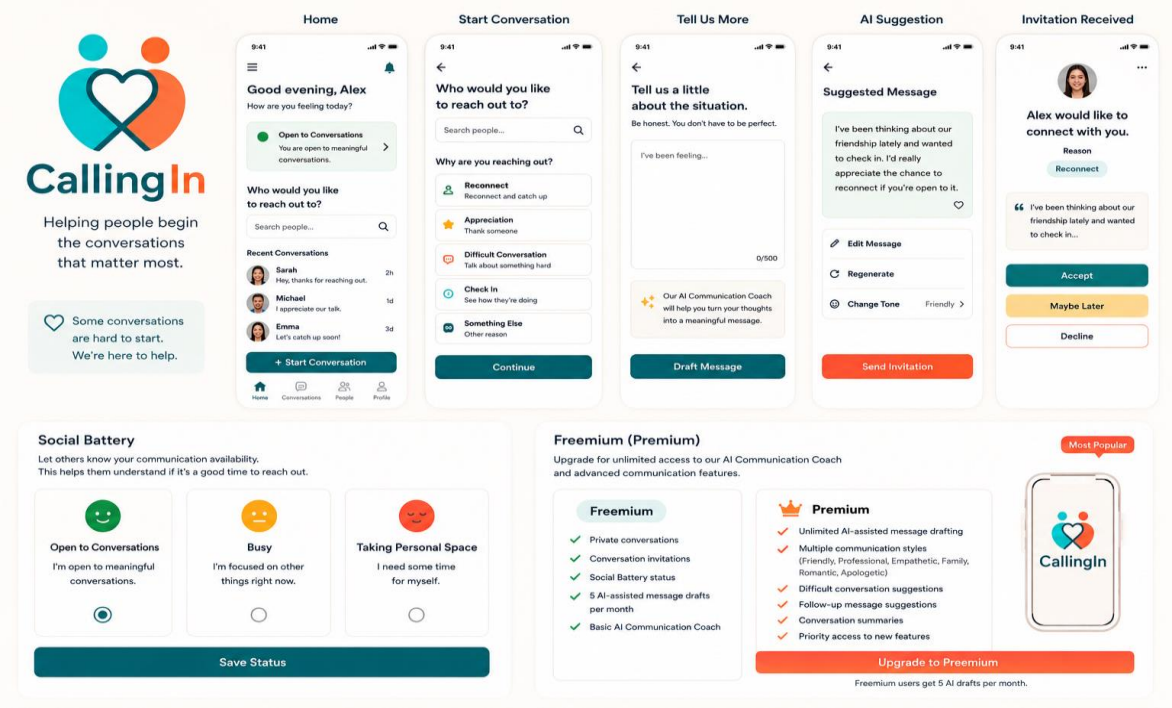
7. The recipient chooses:

• Accept

• Maybe Later

• Decline

8. If accepted, a private conversation begins.



(Figure 1. CallingIn prototype illustrating the proposed user interface, conversation initiation workflow, AI Communication Coach, invitation flow, Social Battery feature, and premium functionality.)

=== Minimum Viable Product (MVP) ===

The MVP for CallingIn will include:

- User registration and authentication
- User profiles
- Conversation invitations
- AI-assisted message drafting
- AI message refinement
- Accept / Maybe Later / Decline invitation workflow
- Basic private conversation page after an invitation is accepted
- Basic Social Battery status

=== Stretch Goals ===

- Real-time messaging
- Typing indicators
- Read receipts
- Conversation summaries

5. Technology Stack

=== Frontend ===

- React *(primary framework. Angular may be evaluated during the project as an alternative framework.)*
- React Hooks
- HTML5
- CSS3
- JavaScript (ES6+)
- Fetch API

=== Backend ===

- Node.js
- Express.js

=== Database ===

- MongoDB

- MongoDB Native Node.js Driver

=== Authentication & Security ===

- Passport.js
- Express Session
- Password Hashing (e.g., bcrypt or the appropriate Passport strategy)

=== AI Integration ===

- OpenAI API (or another approved Large Language Model API)

=== Development Tools ===

- Git
- GitHub
- Visual Studio Code
- Postman (or another API testing tool)
- ESLint
- Prettier

6. Semester Timeline and Syllabus

=== Meeting Schedule ===

Biweekly meetings with the faculty advisor throughout the Fall 2026 semester.

Additional meetings may be scheduled as needed.

❖ Weeks 1–2

Objectives

- Review literature
- Finalize system requirements
- Create GitHub repository

Deliverables

- Approved proposal. Initial GitHub repository

❖ Weeks 3–4

Objectives

- Design system architecture
- Create UI mockups
- Design database schema

Deliverables

- Architecture diagrams
- Wireframes
- Database design

❖ Weeks 5–6

Objectives

- Develop backend. Authentication. Database implementation.

Deliverables

- Backend API
- Authentication
- Database integration

❖ Weeks 7–8

Objectives

- Develop frontend. User authentication. Conversation request workflow.

Deliverables

- Functional frontend
- Connected backend

❖ **Weeks 9–10**

Objectives

- Integrate AI Communication Coach
- Implement prompt generation
- Test AI features

Deliverables

- AI-assisted communication module

❖ **Weeks 11–12**

Objectives

- Testing. Bug fixes. Performance improvements

Deliverables

- Beta version

❖ **Weeks 13–14**

Objectives

- Finalize MVP
- Evaluation
- Presentation
- Final Reflection (Due Friday before Finals Week)

Deliverables

- Final report
- Final presentation & Reflection document

7. Deliverables

- Project Proposal
- Literature review
- GitHub repository
- System architecture
- Minimum Viable Product (MVP)
- CallingIn web application
- Final report
- Project presentation
- Reflection

8. GitHub Repository

A GitHub repository will be created at the beginning of the semester to track source code, documentation, project milestones, and weekly progress. The repository will be shared with the faculty advisor for ongoing review and feedback.

9. Final Reflections

A written reflection will be submitted on the Friday before Finals Week. The reflection will summarize the project's outcomes, challenges encountered, lessons learned, and potential future enhancements for CallingIn.

10. References

References will include peer-reviewed research articles, conference publications, books, and technical documentation related to:

- Generative AI
- Human-Computer Interaction

- Artificial Intelligence
- Digital Communication
- Large Language Models
- Prompt Engineering
- Ethical AI
- OpenAI Documentation
- MongoDB Documentation
- React Documentation
- Node.js Documentation
- MDN